1. What is the payload per message of SIGFOX?

a. 10 bytes
b. 11 bytes
c. 12 bytes
d. 13 bytes

Answer: (c)

Justification: Please refer to Slide 5 of Lecture 1 of Week 7

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2. Which of the following is a processor?

a. DHT
b. LoRa
c. NodeMCU
d. None of the above

Answer: (c)

Justification: Please refer to Slide 7 of Lecture 1 of Week 7

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3. “We can use Arduino IDE to program a NodeMCU board.” --- Is this statement True/False?

a. True
b. False

Answer: (a)

Justification: Please refer to Slide 15-20 of Lecture 1 of Week 7

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4. Which of the following topology(ies) is/are supported by ZigBee?

a. Star
b. Tree
c. Mesh
d. All of the above

Answer: (d)

Justification: Please refer to Slide 5 of Lecture 2 of Week 7

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5. Which software is used to configure XBee?

a. XCTU
b. Arduino IDE
c. NodeMCU
d. DHT

Answer: (a)

Justification: Please refer to Slide 8 of Lecture 2 of Week 7

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6. Which Python library is necessary/imported while using Xbee?

a. numpy
b. TensorFlow
c. DigiMesh
d. OpenCV

Answer: (c)

Justification: Please refer to Slide 11, 12 of Lecture 2 of Week 7

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7. Which of the following statements is true about Complex Event Processing (CEP)?

a. It analyses and correlates event stream from different data sources
b. Can process data in near real-time
c. Extract causal and temporal patterns using predefined rules
d. All of the above

Answer: (d)

Justification: Please refer to Slide 4, 5 of Lecture 3 of Week 7

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8. Edge provides _____ computing, and cloud provides _____ computing.

a. distributed, decentralized
b. centralized , distributed

c. distributed, centralized
d. decentralized , distributed

Answer: (c)

Justification: Please refer to Slide 6 of Lecture 3 of Week 7

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9. What are the Enablers of Prescriptive analytics in the context of IIoT - Processing & Analytics?

a. Optimization/Simulation/Decision models
b. Data mining/Web mining/predictions
c. Dashboard/Reports/Scorecards
d. All of the above

Answer: (a)

Justification: Please refer to Slide 9 of Lecture 3 of Week 7, or page 244 of the book," Introduction to Industrial Internet of Things and Industry 4.0".

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10. What are the component of FarmBeats?

a. Soil sensors, camera, UAVs, weather station, IoT gateway, IoT base station, cloud-services
b. Farmers, Brokers, customers, transport, markets
c. Sensors, weather station, gateway, base station, cloud
d. Sensors, virtual entity, analytics & learning, data management, service management

Answer: (a)

Justification: Please refer to Slide 2 of Lecture 4 of Week 7

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11. What is SWAMP?

a. Water management system
b. Data-driven precision agriculture
c. Irrigation management for different types of crops & climate in different countries
d. Analytics to empower understanding of plant growth according to soil and climatic conditions

Answer: (c)

Justification: Please refer to Slides 4 of Lecture 4 of Week 7

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12. Which of the following provides Cognitive Manufacturing?

a. SmartSantander
b. iRobot-Factory
c. FarmBeats
d. None of the above

Answer: (b)

Justification: Please refer to Slide 14 of Lecture 4 of Week 7

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13. What are the three different phases of smart manufacturing?

a. Organize & process data, decision making, and Generating & controlling the events in real-time
b. Collection of the service offerings & infrastructure, virtualization, allocation & management of services, and on-demand service provisioning
c. Authentication, Authorization and Accounting (AAA), Test bed management, and Application
d. Integration of data and contextual information, synthesis & analysis, innovation in process & production

Answer: (d)

Justification: Please refer to Slide 17 of Lecture 4 of Week 7

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14. What is the three user entities in Industrial Manufacturing?

a. Providers, manufacturers, operators
b. Providers, consumers, operators
c. Providers, manufacturers, distributers
d. Providers, consumers, distributers

Answer: (b)

Justification: Please refer to Slide 21 of Lecture 4 of Week 7

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15. How many major components comprise Industrial Control Systems?

a. 4
b. 5
c. 6
d. 7

Answer: (a)

Justification: Please refer to Slides 2 of Lecture 5 of Week 7

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